PC104 connectors

J7
PC104_H1_H2

H1

Vsys	103	Vsys	PGND 104
	101	Vsys	PGND 102
	99	Vsys	PGND 100
+3.3V SP6	97	SP6	PGND 98
	95	+3.3V	PGND 96
	93	+3.3V	PGND 94
	91	+3.3V	PGND 92
+5V SP5	89	SP5	PGND 90
	87	+5V	PGND 88
	85	+5V	PGND 86
	83	+5V	PGND 84
Vsys SP4	81	SP4	PGND 82
	79	Vsys	PGND 80
	77	Vsys	PGND 78
+3.3V SP3	75	SP3	PGND 76
	73	+3.3V	PGND 74
	71	+3.3V	PGND 72
	69	+3.3V	PGND 70
+5V SP2	67	SP2	PGND 68
	65	+5V	PGND 66
	63	+5V	PGND 64
	61	+5V	PGND 62
Vsys SP1	59	SP1	PGND 60
	57	Vsys	PGND 58
	55	Vsys	PGND 56
	53	Vsys	PGND 54

PGND

H2

RBF_A	51	RBF_A	RBF_B 52
DEP_SW_A	49	DEP_SW_A	DEP_SW_B 50
SP13	47	SP13	SP14 48
SYS_FAULT_n	45	SYS_FAULT_n	WDT_KICK 46
SP11	43	SP11	SP12 44
ENA_5V	41	ENA_5V	ENA_3V3 42
SP9	39	SP9	SP10 40
PPS	37	PPS	DGND 38
SP7	35	SP7	SP8 36
I2C_B_SCL	33	I2C_B_SCL	DGND 34
I2C_B_SDA	31	I2C_B_SDA	DGND 32
I2C_A_SCL	29	I2C_A_SCL	DGND 30
I2C_A_SDA	27	I2C_A_SDA	DGND 28
CAN_B_L	25	CAN_B_L	DGND 26
CAN_B_H	23	CAN_B_H	DGND 24
DGND	21	DGND	DGND 22
CAN_A_L	19	CAN_A_L	DGND 20
CAN_A_H	17	CAN_A_H	DGND 18
DGND	15	DGND	DGND 16
VREF AGND	13	AGND	AGND 14
	11	VREF+	AGND 12
THM6	9	THM6	THM7 8
THM4	7	THM4	THM5 6
AGND	5	AGND	AGND 4
THM2	3	THM2	THM3 2
THM0	1	THM0	THM1 1

DGND
AGND
AGND
AGND
THM7
THM5
AGND
THM3
THM1

notes:
make sure u are using the latest PC104
Please use the same net label neames as used here
SP = spare

3v3 CAN transceiver

U4
TCAN332
+3.3V
CAN_A_TX 1 TXD
CAN_A_RX 4 RXD
VCC 3
GND 2
CANH 7
CANL 6
CAN_A_H
CAN_A_L
C1 100nF
R7 120
JP1 CAN_TERM
DGND
DGND
Bridge JP1 only if this board is at an end of the CAN_A bus

U3 BNO055 Eval board

+3.3V
IMU_RST 6 RST
I2C_A_SDA 4 SDA
I2C_A_SCL 5 SCL
VIN 1
3VO 2
PS0 10
PS1 9
INT 8
IMU_INT
ADR 7
GND 3
DGND
BNO055 in I2C mode (PS0/PS1 low on breakout), addr 0x28 (ADR low). No I2C pull-ups on this board (remove the breakout's 10k SDA/SCL pull-ups).

BOOT_button

+3.3V
SW_Push SW2
BOOT0_sw
R4 10k
BOOT0
R5 2k2
DGND

Reset_button

DGND
SW_Push SW3
C4 100nF
NRST
DGND

SWD headers

+3.3V
J6
SWDIO 4
SWCLK 3
2
1
header_SWD

Power flags (rails come from the PC/104 stack / USB)

+3.3V +5V Vsys VREF VBUS PWR_FLAG PWR_FLAG PWR_FLAG
PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG PWR_FLAG
PWR_FLAG DGND AGND PGND

USB-C Receptacle

J5
TP10
VBUS A4
CC1 A5
CC2 B5
D− A7
D− B7
D+ A6
D+ B6
SBU1 A8
SBU2 B8
GND A12
SHIELD S1
16P_USB_C
5k1 R13
5k1 R14
VBUS
U5
USB_D+ 1
USB_D− 3
6 USBR_D+
4 USBR_D−
2
USBLC6−2SC6
DGND
USBR_D− USBR_D+
ESD prot

MCU supply decoupling — place each 100nF at its VDD/VSS pin pair

+3.3V
C2 10uF
C3 4.7uF
C5 100nF
C6 100nF
C7 100nF
C8 100nF
C9 100nF
C10 100nF
C11 100nF
C12 100nF
C13 100nF
C14 100nF
C15 100nF
C16 100nF
C17 100nF
DGND
C2 10uF at the PC/104 +3.3V entry, C3 4.7uF at the MCU corner feed.
100nF at each VDD/VSS pin pair: C5 (38/39) C6 (51/52) C7 (61/62) C9 (83/84) C10 (94/95) C12 (120/121) C13 (130/131) C15 (16/17), C17 on VBAT. C8 C11 C14 C16 on the four corner feeds.
PCB: all VDD pins tie to a +3.3V copper island under U2, fed through the package corners.

VDDA / VREF+ filter
+3.3V FB1 600R@100MHz
PWR_FLAG
VDDA
C18 1uF
C19 100nF
DGND DGND

Core LDO (VCAP) caps
VCAP1 C20 2.2uF
VCAP2 C21 2.2uF
DGND DGND

Clocks

Y1
25MHz
HSE_IN 1
3 HSE_OUT
2
C22 10pF
C23 10pF
DGND DGND DGND
HSE: 25 MHz, CL = 8 pF
(C = 2 x (CL − 3 pF stray))

Indicators

+3.3V
R1 2k2
D1 PWR
DGND
LED_STAT
R2 2k2
D2 STAT
DGND

U2 STM32H753ZITx

+3.3V
6 VBAT
17 VDD
30 VDD
39 VDD
52 VDD
62 VDD
72 VDD
84 VDD
108 VDD
121 VDD
131 VDD
144 VDD
95 VDD
33 VDDA
VDD33_USB
NRST 25 NRST
BOOT0 138 BOOT0
VDDA 32 VREF+
+3.3V 143 PDR_ON
HSE_IN 23 PH0
HSE_OUT 24 PH1
PA0 34
PA1 35
PA2 36
PA3 37 PPS
PA4 40
PA5 41
PA6 42
PA7 43
PA8 100
PA9 101
PA10 102
PA11 103 USB_D−
PA12 104 USB_D+
PA13 105 SWDIO
PA14 109 SWCLK
PA15 110
PB0 46
PB1 47
PB2 48
PB3 133
PB4 134
PB5 135 IMU_RST
PB6 136
PB7 137
PB8 139
PB9 140
PB10 69 I2C_A_SCL
PB11 70 I2C_A_SDA
PB12 73 CAN_A_RX
PB13 74 CAN_A_TX
PB14 75
PB15 76
PC0 26
PC1 27
PC2_C 28
PC3_C 29
PC4 44
PC5 45
PC6 96
PC7 97
PC8 98
PC9 99
PC10 111 LED_STAT
PC11 112
PC12 113
PC13 7
PC14 8
PC15 9
PD0 114 IMU_INT
PD1 115
PD2 116
PD3 117
PD4 118
PD5 119
PD6 122
PD7 123
PD8 77
PD9 78
PD10 79
PD11 80
PD12 81
PD13 82
PD14 85
PD15 86
PF0 10
PF1 11
PF2 12
PF3 13
PF4 14
PF5 15
PF6 18
PF7 19
PF8 20
PF9 21
PF10 22
PF11 49
PF12 50
PF13 53
PF14 54 I2C_B_SCL
PF15 55 I2C_B_SDA
PE0 141
PE1 142
PE2 1
PE3 2
PE4 3
PE5 4
PE6 5
PE7 58
PE8 59
PE9 60
PE10 63
PE11 64
PE12 65
PE13 66
PE14 67
PE15 68
VCAP1 71
VCAP2 106
VCAP
VCAP
VSS 16
VSSA 31
DGND

STM32H753 pin map (set the AF in firmware):
PA3 PPS TIM2_CH4 input capture
PF14/PF15 I2C_B I2C4 SCL/SDA
PB10/PB11 I2C_A I2C2 SCL/SDA
PB12/PB13 CAN_A FDCAN2 RX/TX via U4
PB10/PB11 BNO055 on I2C_A (I2C2), PB5 IMU_RST, PD0 IMU_INT
PA11/PA12 USB OTG_FS (no VBUS sensing), PA13/PA14 SWD, PC10 LED_STAT

STM32H753ZI + BNO055; CAN_A, I2C_A, I2C_B and PPS on the PC/104 stack
Team Anant — 3U EM

Sheet: /		
File: main_ADCS.kicad_sch		
Title: AnantSAT−II ADCS board		
Size: A3	Date: 2026−09−27	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/1